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Efference-copy gain, gating, and current self-state in a delayed fly steering model

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Abstract

Flying insects separate visual motion caused by their own turns from motion caused by the world, often via an efference copy. Two model families implement this idea. Graded models subtract a continuous prediction of self-generated optic flow; all-or-none models suppress the stabilizing optomotor pathway while an object-driven turn is underway. We extend a published fly-steering model with an explicit transport delay and compare five core variants with command-copy and goal-feedforward controls across efference-copy gains, gate thresholds, velocity-nonlinearity shapes (clip, smooth, unclipped), simulation windows, and adaptive update rules. Three results survive these stress tests. First, gain. At the source model's fixed gain of 0.05, fixed-gain graded cancellation is inert and indistinguishable from no efference copy. At gains of 0.25 or more, delayed-sensory cancellation is harmful at 50 ms, failing in 100% of noiseless runs versus 75% for no efference copy. Second, gating. At delays of 20 to 50 ms, gating on the filtered plant state prevents failure (0%) at every gate threshold from 0.01 to 0.5 of the bar-torque scale and, separately, across all three velocity-nonlinearity shapes, while the continuous models fail in 62.5% to 100% of runs; the second gating signal in the threshold sweep matches these zeros except one

corner cell (threshold 0.5, 50 ms, clip: 25%). The advantage is not asymptotic. At 100 ms it depends on the velocity-nonlinearity shape, and at 140 ms or longer it is not established. Third, adaptation. Neither the source-style squared-error update nor a signed-gradient update rescues failure at 50 ms. The signed update keeps the learned gain lower (0.22 to 0.40 versus 0.51 to 1.05), but failure remains 100%. A pre-plant-filter command copy also did not rescue delay; only a current-visual oracle reduced failure, confirming that the earlier long-delay benefit was an oracle effect. A goal-based feedforward branch that uses current proprioception does rescue: gating with a feedforward weight of 0.5 or more fails in no parameter set through 200 ms at both 6 and 10 s windows, and the rescue disappears when that branch's self-state estimate is delayed. The results identify efference-copy gain and short-delay gating as robust variables, show that earlier long-delay gating claims were artifacts of a 2.5 s window and the velocity-nonlinearity clip, and show that what a delayed visual loop lacks in this architecture is current self-state rather than a copy of the issued command. The model is a stylized reimplementation. It does not reproduce the source's zero-delay ranking and makes no claim about the mechanism used by the fly.

Keywords: efference copy, reafference, optic flow, sensorimotor delay, gating, Drosophila, flight steering

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Introduction

An animal that moves through the world creates much of the sensory input it receives. The visual flow produced by a fly's own turn is indistinguishable, at the retina, from the flow produced by an external rotation. Behavioral and physiological work has established that flies solve this problem in part by combining visual signals with motor-related signals. Proprioceptive feedback sets the gain of visual steering responses in flying Drosophila (Bartussek and Lehmann, 2016), and visual and haltere mechanosensory angular-velocity signals are summed in flight control (Sherman and Dickinson, 2004). During voluntary turns, horizontal-system cells in the lobula plate receive motor-related inputs with a sign and latency appropriate to suppress the cell's response to the turn (Kim et al., 2015). A visual cell type in the lobula, LT52, receives non-visual feedback about limb movement and is positioned to suppress steering toward the fly's own leg during grooming (Hartman et al., 2025); that limb-movement feedback reaches a visually guided steering circuit illustrates the general point that non-visual self-motion signals shape turning. The computation is therefore not purely visual. It depends on an internal prediction of the sensory consequences of action, the classical reafference principle (von Holst and Mittelstaedt, 1950).

How that prediction is used is less clear. One class of models subtracts a graded, continuous estimate of self-generated optic flow from the visual drive. A second class gates the stabilizing pathway off while an object-driven turn is in progress. Canelo et al. (2024) compared an addition-only model, a graded efference-copy model, and an all-or-none model against behavioral data from tethered flying flies. The all-or-none model accounted for the observed invariance of object-evoked turns across sparse and dense backgrounds; graded cancellation and simple addition did not. That result favors gating under the conditions tested.

The comparison left a major physical constraint out of the main analysis, namely delay. Any visuomotor loop has a delay between a motor command and the visual feedback it produces, and between the visual signal and the motor response. The source model included a delay block only in a supplementary figure and omitted it from the main simulations. In the source code, that supplementary comparison varies whether the torque is passed through the tau filter, not a transport delay on the visual feedback. Our delay is therefore a different manipulation. This omission matters because delayed negative feedback is a classic route to oscillation and loss of stability, with a classical control literature on compensation by internal forward models that predict current state rather than replay delayed measurements (Smith-predictor-style architectures). The delays involved are not hypothetical: motion-sensitive visual neurons in the fly respond with latencies of tens of milliseconds that vary an order of magnitude with stimulus and physiological conditions (Warzecha and Egelhaaf, 2000). A graded cancellation signal that is itself delayed can add a second delayed loop to the system. A binary gate can remove the loop while a turn is underway. The two strategies may therefore rank differently when delay is treated as a first-class variable.

This paper asks a narrow question. Under an explicit transport delay, which variants preserve bounded steering, and how do the answers depend on efference-copy gain, gate threshold, V-term treatment, and simulation window? We do not claim to identify the fly's mechanism. We use the model as a computational assay that generates testable predictions for delayed-feedback experiments.

We also report two corrections. An earlier analysis claimed that gating was robust through 100 ms and that linear stability explained a large-signal effect. The former was a finite-window and V-clip artifact. The latter was based on unreliable

delay-eigenvalue tracking. Both claims are removed. The remaining results are narrower, but they are stable across the stress tests reported below.

Results

A minimal steering model and five variants

We used the flight-simulator equations of Canelo et al. (2024) as the starting point. The fly's heading error is corrected by a bar pathway that responds to the angular position and velocity of a target, and by a grating pathway that responds to wide-field background motion. The body is a second-order yaw system with moment of inertia `I` and drag `beta`. The bar torque is low-passed with `tau_bar = 0.05 s`, the grating torque with `tau_opto = 0.04 s`, and the grating gain is `k_opto = 25 x 10^-12 N m` per rad/s.

We introduced a delay `d` by evaluating the visual state at time `t - d`: the bar angle and its derivative, the grating angle and its derivative, the heading, and the angular velocity. The body state and the torque low-pass filters remain at the current time. This is a deliberately simple transport delay. It is not a full delay differential equation treatment, but it captures the essential loop: visual information about the current state arrives late. The heading and angular velocity are treated as visually sensed body states; delaying them is an assumption, not a claim about proprioception.

We compared five variants:

- **addition:** bar and grating torques are summed; no efference copy.
- **graded filtered:** a fixed-gain internal prediction `omega_ec` is driven by the filtered bar-torque state `F_bar` and subtracted from the grating drive.
- **graded unfiltered:** the same graded architecture, but `omega_ec` is driven by the instantaneous bar-pathway output computed from the current visual state.
- **gate filtered:** the grating gain is set to zero when the filtered bar torque exceeds a threshold.
- **gate unfiltered:** the grating gain is set to zero when the instantaneous bar-pathway output exceeds a threshold.

The labels "filtered" and "unfiltered" are deliberate. The unfiltered variants are not copies of an issued motor command. They are bar-pathway signals computed from the current visual state, and they therefore differ from the filtered variants in both timing and filtering. The comparison tests the effect of signal timing and filtering, not a sensory-versus-motor distinction.

All variants share the body, sensory time constants, bar and grating stimuli, integration step, and failure criterion. The only differences are the efference-copy architecture and the signal used by the gate.

Efference-copy gain: the source's fixed gain is inert; higher delayed-sensory gain is harmful

We swept the graded efference-copy gain from 0 to 1.0 at delays of 0.02, 0.05, 0.10, and 0.20 s across 12 parameter sets (1,440 simulations; Fig. 2). The source's fixed graded implementation uses a gain of 0.05 on the bar torque in its Figure 5 code, so we treat 0.05 as the reference gain.

At gain 0 or 0.05, both graded variants were indistinguishable from addition at every delay. At 50 ms, all three failed in 75% of parameter sets. At gains of 0.25 or more, delayed-sensory cancellation became harmful. Failure at 50 ms rose to 100% (12/12), while addition remained at 75% (9/12). At gains of 0.75 or 1.0, failure at 20 ms also rose, from 42% to 50%.

The unfiltered graded variant behaved differently at the short 2.5 s window. At gains of 0.5 or more, its 200 ms failure was 25% to 42%. At unity gain, extending the window to 6 s raised the 200 ms failure to 62.5%. We therefore do not treat it as a robust long-delay advantage.

These regimes are not a monotonic effect of gain. A strong delayed-sensory estimate adds a delayed feedback loop, which destabilizes the system. A strong unfiltered drive can provide a lead signal at long delays, but it also overshoots at short delays. The source's fixed gain of 0.05 sits in the inert regime. It is too weak to either help or hurt. This is the first main result: the effect of graded cancellation is not a property of "graded versus gating" alone. It depends on the gain and on the filtering of the signal.

Gating prevents failure at short delays

We next asked whether gating prevents failure at delays where the continuous models fail. At a 6 s window, with eight parameter sets and two noise seeds, gating was perfect at 20 and 50 ms. `gate filtered` and `gate unfiltered` both failed in 0 of 16 runs at 20 and 50 ms. Addition failed in 75% and 100% of runs at those delays, graded filtered in 87.5% and 100%, and graded unfiltered in 75% and 62.5% (Fig. 3).

At 100 ms, the picture changed. `gate filtered` failed in 0% of runs at 2.5 and 6 s, but in 18.8% at 10 s. `gate unfiltered` failed in 0%, 6.2%, and 6.2% at the same windows. Addition and graded filtered failed in 100% of runs at 100 ms in every window; graded unfiltered failed in 62.5%, 75%, and 75%. At 140 ms, `gate filtered` failed in 18.8%, 68.8%, and 75% at 2.5, 6, and 10 s, while `gate unfiltered` failed in 75% at all three windows. At 200 ms, both gating variants failed in 75% of runs at the 6 s window, and addition and graded filtered failed in 100%.

The defensible claim is therefore narrow. Gating prevents the early failures that continuous models show at 20 to 50 ms. It buys time at 100 ms, but the advantage is small and window-dependent. It does not provide an asymptotic stability advantage at 140 ms or longer.

The long-delay gating claim was a window artifact

An earlier version of this analysis used a 2.5 s window and reported that `gate filtered` failed in only 3 of 16 runs at 140 ms. That number was a window artifact. When the same 16 runs were integrated to 6 s, 11 failed. At 10 s, 12 failed. `gate unfiltered` failed in 12 of 16 at all three windows. The 140 ms comparison therefore does not support a gating advantage (Fig. 3).

The same window check changes the interpretation of the 200 ms critical delay. At 200 ms, `gate filtered` failed in 75% of runs at 6 s, and `gate unfiltered` also failed in 75%. The earlier median of 200 ms was a censored first-failure time, not a stability boundary. We no longer report a 200 ms critical delay for gating.

The 100 ms advantage depends on the V-term treatment and gate threshold

We tested whether the short-delay gating advantage is an artifact of the bar-velocity V term. The source's V term becomes negative for heading errors above `sqrt(10)` rad, which is inside the large-excursion regime that determines failure. We compared the original hard clip (`max(0, 20 − 2 error^2)`), a smooth saturating form (`20 / (1 + (error/3.162)^4)`), and the unclipped source form at a 6 s window (Fig. 4, left).

At 20 and 50 ms, the failure rates were identical across the three V treatments. Gating remained at 0%, and the continuous models remained at 62.5% to 100%. At 100 ms, the treatments diverged. `gate filtered` failed in 0% with the clip, 75% with the smooth form, and 0% with the unclipped form. `gate unfiltered` failed in 6.2%, 75%, and 6.2%. At 140 and 200 ms, failure was 62.5% to 100% under all three treatments; the lowest cell is graded unfiltered at 200 ms (62.5% in all three), and `gate filtered` at 140 ms is 68.8% under clip and unclipped.

We also swept the gate threshold from 0.01 to 0.5 of the bar-torque scale at the same 6 s window, for both gating signals. For `gate filtered`, gating remained at 0% across every threshold and both the clip and smooth V treatments at 20 and 50 ms. The second signal, `gate unfiltered`, matched these zeros everywhere at 20 and 50 ms except one corner: threshold 0.5 under clip, where it failed in 25% of runs at 50 ms (and in 62.5% at 100 ms, versus 37.5% for `gate filtered`). At 100 ms for `gate filtered`, the clip treatment produced 0% failure for thresholds below 0.5 and 37.5% at 0.5, while the smooth treatment produced 75% at every threshold. At 140 ms, every threshold and treatment produced 75% failure.

The short-delay gating advantage is therefore robust to gate threshold and V treatment. The 100 ms advantage is not. It requires the hard clip and a threshold below 0.5, and it disappears under a smooth V term.

Adaptive updates do not rescue delay

An earlier analysis used the source's squared-error update and concluded that adaptation learned a higher gain and failed. The source update uses a non-negative squared error, so the gain can only increase. We therefore compared three update rules at a 6 s window: the source-style squared error, a signed-error gradient, and a signed-error gradient with the opposite sign (Fig. 5).

The source-style update drove the mean gain from 0.51 at 20 ms to 1.05 at 200 ms. Failure was 50% at 20 ms and 100% at 50, 100, and 200 ms. The signed gradient kept the mean gain lower, from 0.22 at 20 ms to 0.40 at 200 ms. Failure was

unchanged: 50% at 20 ms and 100% at the longer delays. The opposite-sign control also left failure unchanged, with a mean gain of 0.11 to 0.38.

Adaptation therefore does not rescue bounded steering at 50 ms in this architecture, whether the gain is allowed to rise monotonically or is updated with a signed error. The learned gain changes, but the failure boundary does not. We do not claim that all adaptive controllers fail. We claim that this MLP, with this oracle bar-only target, does not rescue the delay problem. To check that the negative result is not an artifact of updating the weights between solver steps, we re-implemented the update by integrating the weights as states of the differential equation with a rate matched to the discrete step. Across 96 paired runs (four delays, two update rules, twelve parameter sets), the two forms agreed on failure classification and failure time in every pair. Before the runs diverge, trajectory differences stay at floating-point scale in most pairs, with exceptions up to 0.21 rad (source rule) and 4.1 rad (gradient rule) at 20 ms, where the runs themselves diverge anyway; none of these prefix differences changed a failure classification.

The source's zero-delay ranking is not reproduced

The source's behavioral result is motivation for comparing the model families, not a validation of our implementation. In a zero-delay check with our bar and grating step, the ratio of peak heading at high versus low grating gain was 0.9148 for addition, 0.9830 for gate filtered, 0.9996 for graded unfiltered, and 0.9998 for graded filtered. Graded cancellation was at least as invariant to grating gain as gating in this setting, which does not reproduce the source's ranking.

We also ran a check with the source's scaled parameters, a constant grating angle, and grating gains of 0.05 and 0.7. At a near-zero gate threshold, gate filtered was invariant (59.956 versus 59.956 degrees) by construction, because the gate is effectively always closed. At the main threshold used in this paper, gate filtered was about 4% sensitive (59.890 versus 57.422 degrees). Graded cancellation at a gain of 0.5 was about 0.3% sensitive (59.923 versus 59.770 degrees), and addition was 15% sensitive (59.846 versus 50.846 degrees). The source's ranking is therefore not recovered at a meaningful effect size. The model is a stylized two-pathway assay inspired by the source, not a faithful extension of it.

A true command copy does not rescue delay

To separate a command copy from plant-state feedback, we added a pre-plant-filter command signal `u_t`, computed from the same delayed visual estimate that drives the controller, and used its copy for the efference copy or gate. At a 6 s window and gain 1.0, graded command and graded filtered had identical failure rates: 87.5% at 20 ms and 100% at 50, 100, 140, and 200 ms (Fig. 6, left). Gate command and gate filtered also had identical failure rates: 0% at 20, 50, and 100 ms and 75% at 140 and 200 ms. The command gate had a lower duty cycle (0.35, 0.65, 0.79, 0.82, 0.85) than the filtered gate (0.44, 0.78, 0.85, 0.87, 0.87), but this did not change failure (Fig. 6, right). The only variant with lower failure at long delays was the old unfiltered oracle, which uses the current visual state (75%, 62.5%, 75%, 75%, and 62.5% at 20 to 200 ms). Its apparent benefit is therefore an oracle effect, not a motor-copy effect. The source question closes: in this architecture, a true command copy does not rescue delay. The graded-family identity and the gate-goal zeros reproduce identically in all three seeds of a multi-seed re-run (24 parameter-seed combinations per cell). The gate pair is not identical at the run level: `gate filtered` drifts by one or two runs across seeds (0%, 0%, 12.5% at 100 ms; 75%, 62.5%, 62.5% at 140 ms) while `gate command` is 0% and 75% in every seed, a drift that does not favor the command copy.

A goal feedforward rescues delay, and it requires current self-state

A command copy carries no information that is absent from the delayed visual estimate, so we added a branch that does. The goal feedforward computes a torque from an instructed goal trajectory (a first-order approach to 60 degrees starting at 0.5 s with time constant 0.1 s) and the current proprioceptive state (heading and angular velocity). The torque is added to the bar pathway with weight `ff_gain` from 0 to 1, and the same signal is used by the graded efference copy or the gate. At `ff_gain = 0`, graded goal is identical to graded filtered.

The graded family improves only partially. At `ff_gain = 1.0`, failure is 75%, 62.5%, 75%, 75%, and 75% at 20 to 200 ms, matching the current-visual oracle at four of five delays (they differ at 200 ms: 75% versus 62.5%, one of eight parameter sets); the delayed bar loop is unchanged. At 0.75 it is 75%, 75%, 87.5%, 87.5%, 75%; at 0.5 it is 75%, 87.5%, 100%, 100%, 100% (Fig. 7b).

The gated family rescues completely. Gate goal at `ff_gain` of 0.5 or higher has 0% failure at every delay from 20 to 200 ms, and this holds at both 6 s and 10 s windows and in all three seeds of a separate multi-seed re-run (Fig. 7a). In the feedforward window check, `gate filtered` fails in 0% of runs at 100 ms at 6 s and 12.5% at 10 s (1 of 8; a separate

window check with 16 runs gives 18.8%), while failing in 75% of runs at 140 and 200 ms at 6 s. At `ff_gain = 0.25` the gate is partial: 0%, 0%, 25%, 25%, 37.5%. The gate duty cycle falls as `ff_gain` rises (from 0.44/0.78/0.85/0.87/0.87 at 0 to 0.31/0.58/0.62/0.64/0.67 at 1.0), because the feedforward term shares the load.

The result does not require an exact instruction. With goal amplitudes of 45, 60, and 75 degrees and time constants of 0.05, 0.10, and 0.20 s, gate goal at `ff_gain = 1.0` keeps 0% failure at 50, 100, and 200 ms for all nine combinations; at 0.5 only the 45-degree undershoot at 200 ms fails, at 12.5%. Graded goal failure rates are unchanged by mismatch (0.625 to 0.75), because that family is limited by the delayed bar loop rather than by the goal.

Two controls identify what the feedforward branch is actually using. First, delaying the goal itself by 0 to 200 ms changes nothing: gate goal at `ff_gain = 1.0` stays at 0% failure for goal delays of 0, 50, 100, and 200 ms at visual delays of 100 and 200 ms, and graded goal stays at 75% (Fig. 7d). Second, replacing the current proprioceptive state in the feedforward branch with the same delayed values used by the visual path removes the rescue: gate goal at `ff_gain = 1.0` rises from 0% to 75% at 100 ms and to 75% to 100% at 200 ms, and graded goal rises from 75% to 100% (Fig. 7c). The delayed-self-state gate is worse than plain gate filtered at 100 ms (75% versus 0%), so a delayed feedforward torque is actively harmful. What the delayed visual loop lacks is current self-state, not a copy of the issued command and not early knowledge of the target.

The rescue extends to 500 ms and survives noise

A 10 ms grid from 0 to 250 ms (6 s window) puts these boundaries on a finer scale (Fig. 9a). The first delay at which failure reaches 50% is 20 ms for graded filtered and 10 ms for graded goal; gate filtered crosses at 130 ms; gate goal at `ff_gain = 1.0` never reaches 50% failure anywhere in the grid. Pushing to 250, 300, 400, and 500 ms at a 10 s window (Fig. 9b), gate goal at 1.0 stays at 0% failure at every delay, while gate goal at 0.5 degrades from 0% at 200 ms to 12.5% at 250 and 300 ms, 62.5% at 400 ms, and 50% at 500 ms; gate filtered fails in 75% of runs throughout, and graded goal is flat at 75%. The rescue therefore has a gain-dependent boundary between 300 and 400 ms in this grid.

Raising the noise standard deviation by two orders of magnitude above the main sweeps, to `1e−11 N m`, changes no conclusion (Fig. 10). The only movement is gate filtered at 100 ms, from 0% at the two lowest noise levels to 12.5% at the two highest; gate goal at 0.5 and 1.0 stays at 0% at 100 and 200 ms at every noise level, and the graded variants are unchanged.

Example trajectories at 200 ms and a 10 s window show the qualitative split directly (Fig. 11). Addition and graded filtered leave $\pm 2\pi$ within about 2 s and grow to 1e8 to 1e13 rad; gate filtered and gate goal remain bounded, ending at 1.27 and 1.11 rad with small oscillations around the 60-degree target. The gates differ in how often they close: the filtered gate is closed in 86% of samples, the goal gate in 25%, because the feedforward term carries part of the load. Failure-free curves at 140 ms over the three-seed run (Fig. 12) show when failures occur: the graded variants fail within the first second, the gate variants that fail do so only after about 3 s, and gate goal never fails. A body-parameter phase diagram over four inertias, three drags, two grating gains, and eight delays (Fig. 13) shows the same asymmetry across the body grid: averaged over body parameters, gate filtered rises from 12.5% failure at 20 ms to 66.7% at 200 ms and beyond, while gate goal at `ff_gain = 1.0` fails in none of the 24 body-gain combinations at any delay from 20 to 400 ms.

The rescue needs fresh self-state, not goal timing or goal content

Three further controls close the remaining gaps in the feedforward construction (Fig. 14).

First, the rescue does not depend on goal-stimulus alignment. Moving the goal onset from 0.3 to 0.8 s relative to a fixed bar onset of 0.5 s changes no failure rate: gate goal stays at 0% at 100 and 200 ms for both `ff_gain` 0.5 and 1.0, and graded goal stays at 75% (Fig. 14b). The instruction need not be temporally aligned with the stimulus it compensates.

Second, goal content is not special. Replacing the goal-tracking torque with pure derivative feedback on the current angular velocity, a damping term that contains no target information, keeps the gate rescue: at 200 ms, damping gains of `1e−11` and `2e−11` give 0% failure, versus 75% for gate filtered and 0% for the goal-tracking reference, and `0.5e−11` gives 50% (Fig. 14c). The graded family improves only partially, from 100% to 50% to 62.5% at the highest damping gain. What the gate needs is a zero-delay stabilizing signal computed from current self-state; whether that signal encodes a goal is irrelevant. Independent experiments point the same way. Antennal mechanosensory input stabilizes a visual groundspeed controller whose processing delay is on the order of 100 ms, and removing the antennae degrades regulation (Fuller et al., 2014), and proprioceptive feedback sets visuomotor gain (Bartussek and Lehmann, 2016). What has not been done, in

any animal, is the experiment prediction 6 proposes: impose a visual feedback delay and vary the latency of the self-state pathway.

Third, the rescue needs a fresh self-state sample. Aging the self-state used by the feedforward branch while holding the visual delay fixed produces a sharp dose-response (Fig. 14a). At a 100 ms visual delay, gate goal stays at 0% for sample ages up to 30 ms and jumps to 75%, the gate filtered level, at 40 ms and beyond. At 200 ms the degradation begins one step earlier, 25% at 20 ms and 75% at 30 ms. Graded goal is at its 75% floor when the sample is fresh and reaches 100% by 10 to 20 ms of age. The self-state pathway must therefore deliver heading and angular velocity estimates roughly 20 to 30 ms old or fresher to sustain the rescue, a number that is directly testable in preparations with measurable proprioceptive or efference-related latencies.

An equal-information observer baseline asks whether any estimate would do. Replacing the self-state sample with a second-order Taylor reconstruction computed from the delayed observation alone, with no additional information, does not substitute for fresh state: gate goal with the predicted state fails in 50% of runs at 50 ms and 100% at 100 ms and beyond, while the current-state version stays at 0% at every delay (Fig. 14; claim-level data in the observer check). The rescue is therefore not a generic information-availability advantage that any trivial estimator would recover. A perfect internal-model observer remains an untested upper bound.

Predictions

The model makes six predictions that can be tested without changing the core architecture.

1. **Short-delay gating.** At feedback delays of 20 to 50 ms, a gating observer should show fewer large heading excursions than a continuous-cancellation observer, across gate thresholds from 0.01 to 0.5 of the bar-torque scale and across V-shaped versus smooth velocity nonlinearities; the second gating signal should match except at the highest threshold in the clip regime.
2. **The 100 ms boundary.** At 100 ms, the gating advantage should depend on the V-shaped velocity nonlinearity. A preparation or model with a smooth saturating velocity response should lose the advantage, while a hard-threshold preparation should retain it.
3. **No long-delay advantage.** At delays of 140 ms or longer, gating should not show a boundedness advantage over continuous cancellation in this architecture.
4. **Gain.** At the source's fixed gain of 0.05, graded cancellation should be inert, indistinguishable from no efference copy. At gains of 0.25 or more, delayed-sensory cancellation should be harmful at 50 ms.
5. **No command-copy rescue.** A pre-plant-filter copy of the issued command should not rescue failure at 50 ms in this architecture, because the command is still based on the delayed visual estimate.
6. **Self-state rescue.** Stabilization under visual delay should survive when a goal-based pathway has access to current self-state, and should fail when that pathway's self-state estimate is delayed as well; a pure efference copy should help neither way. The model puts the required self-state freshness at roughly 20 to 30 ms. In an animal, this translates to a test of whether an imposed visual delay is tolerated when proprioceptive or goal-related pathways are intact, and whether it becomes intolerable when those pathways are delayed to the same degree.

Discussion

The central result is that efference-copy gain is a first-class variable. At the source model's fixed gain of 0.05, graded cancellation is inert. At higher gains, a delayed sensory estimate is harmful. This is not a subtle shift in a continuous metric. It is a change in whether the efference copy participates in the loop at all.

The second result is that gating buys time at short delays, not asymptotic stability. At 20 to 50 ms, gating prevents the early failures that continuous models show. The advantage survives gate thresholds from 0.01 to 0.5 for the filtered plant-state gate, with one corner exception for the second gating signal, and it survives all three V treatments. At 100 ms, it depends on the V treatment. At 140 ms and beyond, it is not established. The earlier claim that gating was robust through 100 ms or 200 ms was a finite-window and V-clip artifact. We report the correction rather than the original claim.

The third result is that adaptation is not a rescue in this architecture. A signed-error gradient keeps the learned gain lower than the source-style update, but failure at 50 ms remains 100%. The negative result is therefore not an artifact of a monotone update. It is a property of the delayed loop with this controller and this oracle target.

Two further results complete the mechanism argument. A pre-plant-filter command copy does not rescue delay; the only variant with a long-delay benefit was the current-visual oracle, so the earlier motor-copy reading was an oracle effect. A goal feedforward, computed from an instructed target and current proprioception, does rescue: gating with `ff_gain` of 0.5 or higher fails in 0% of parameter sets through 200 ms at both 6 and 10 s windows, robust to goal amplitude and time-constant mismatch and to torque noise up to 100 times the main sweeps. Two controls locate the mechanism. Delaying the goal itself by up to 200 ms does not remove the benefit, while replacing current proprioception in the feedforward branch with delayed values does, and makes the gate worse than no feedforward at all. Two further controls remove the construction's remaining privileges: decoupling the goal onset from the stimulus by up to 200 ms in either direction changes nothing, and replacing the goal-tracking torque with pure current-state damping, which contains no target information, keeps the rescue. The rescue does, however, die a sharp death when the self-state sample ages: gate goal degrades between 20 and 40 ms of sample age. In this architecture, the delayed visual loop therefore lacks current self-state, and current self-state alone is enough; a copy of the issued command is not a substitute, and neither is early knowledge of the target. Figure 8 summarizes the three cases; Figure 14 details the freshness dose-response.

This conclusion has a classical flavor. The reafference principle (von Holst and Mittelstaedt, 1950) already distinguishes reafference from exafference, and the forward-model and Smith-predictor traditions hold that delay is best tolerated by predicting current state rather than replaying past input or issued commands (for reviews and extensions see the delayed-feedback and internal-model literature cited in the Introduction; the head-direction-to-steering work of Westeinde et al. 2024 is a recent example in fly navigation). What is new here is not the state-versus-command distinction in the abstract. It is the demonstration, inside a published insect-steering architecture under stress tests, that a copy of the issued command adds nothing while a current-self-state pathway rescues fully, and that the rescue survives goal mismatch and noise but dies when the self-state estimate is delayed.

The relation to the source model is deliberately modest. We do not reproduce the source's zero-delay background ranking. At the main gate threshold, the model is about 4% sensitive to grating gain, while graded cancellation at a gain of 0.5 is about 0.3% sensitive (0.15 degrees of 59.9 degrees). The source's behavioral result is motivation, not validation. The model is a stylized two-pathway assay, not a faithful extension.

The relation to the broader connectome-to-behavior literature is indirect. Recent work has asked whether connectome-constrained networks predict neural activity (Lappalainen et al., 2024) and reproduce behavior (Bates et al., 2026), whether circuit motifs support robust optic flow processing (Erginkaya et al., 2025), whether behavior is sufficient evidence for biological fidelity (Dhawan, 2026; Brunton et al., 2026), whether lesion phenotypes can distinguish biological wiring from degree-matched nulls (Connectome Wiring Shapes Motor Lesion Phenotypes, 2026; Dhiman, 2026), and whether embodied simulations support realistic sensorimotor control (Wang-Chen et al., 2024). Our model is not connectome-constrained. It is a low-dimensional assay of a computational motif. Its value is that it isolates variables that are easy to manipulate in experiment and easy to miss in a static model comparison, namely efference-copy gain, feedback delay, gate threshold, and the shape of the velocity nonlinearity.

The model has clear limits. It is a two-pathway simplification of a circuit that contains many cell types, multiple motion pathways, and nonlinear dendritic processing. The parameters are taken from a published model and not fitted to new data. The delay is a simple transport delay. The failure criterion is a modeling choice. The model has no head or neck dynamics, halteres, or descending motor pathways. It does not include learning or adaptation in the main fixed-gain comparison; the adaptive variant uses an oracle bar-only target, and its step-wise weight update was verified to match an ODE-internal form in failure classification and failure time. It does not reproduce the source's zero-delay ranking. It makes no claim about the mechanism used by the fly.

A linear stability analysis was attempted. The delay-eigenvalue root tracking was unreliable: the reported 300 ms value for one variant was a censoring limit, not a crossing, and the eigenvalue curves were discontinuous. We therefore do not use that analysis. The nonlinear simulations are the primary evidence. A validated delay-eigenvalue solver is future work.

The next step is an experimental test of the self-state proposition (prediction 6), not further analysis of the current variants.

Methods

Model

The model follows the flight-simulator equations of Canelo et al. (2024). The state is the fly heading `psi`, angular velocity `omega`, bar torque `F_bar`, and grating torque `F_grating`. For the graded variants, an additional state `omega_ec` represents the efference-copy prediction.

The body is a second-order yaw system:

```
dpsi/dt = omega
domega/dt = (−beta ∗ omega + F_bar + F_grating) / I
```

The bar pathway is:

```
error = wrap(theta_bar − psi)
P(error) = p_gain ∗ sin(error)
V(error) = v_gain ∗ f_V(error)
dF_bar/dt = (−F_bar + P(error) + (dtheta_bar − omega) ∗ V(error)) / tau_bar
```

where `p_gain = 10e−11` and `v_gain = 1e−12`. The function `f_V` is `max(0, 20 − 2 error^2)` for the clip mode, `20 / (1 + (error/3.162)^4)` for the smooth mode, and `20 − 2 error^2` for the unclipped mode. The clip mode is the default. The source uses the unclipped form, which becomes negative for `|error| > sqrt(10)` rad.

The grating pathway is:

```
dF_grating/dt = (−F_grating + (dtheta_grating − omega) ∗ k_opto) / tau_opto
```

For graded variants, the grating drive becomes `dtheta_grating + ec_gain ∗ omega_ec − omega`, and:

```
domega_ec/dt = (−beta ∗ omega_ec + F_bar) / I              (graded filtered)
domega_ec/dt = (−beta ∗ omega_ec + F_bar_unfiltered) / I   (graded unfiltered)
```

where `F_bar_unfiltered` is the instantaneous bar-pathway output computed from the current visual state. For gate filtered, `k_opto` is set to zero while `|F_bar| > gate_threshold`; for gate unfiltered, the gate uses `|F_bar_unfiltered| > gate_threshold`. The default `gate_threshold` is `1.5e−11 N m`, which is 0.15 of `p_gain`.

For the command-copy variants, the efference copy or gate uses the bar command issued by the controller at the current time, before the plant filter:

```
F_bar_command = P(error_d) + (dtheta_bar_d − omega_d) ∗ V(error_d)
domega_ec/dt = (−beta ∗ omega_ec + F_bar_command) / I   (graded command)
```

where `error_d`, `dtheta_bar_d`, and `omega_d` are the delayed visual values. For gate command, `k_opto` is set to zero while `|F_bar_command| > gate_threshold`. The command copy is pre-plant-filter but still based on the delayed visual estimate.

For the goal feedforward variants, the goal heading follows a first-order approach to the instructed amplitude with time constant `goal_tau_s`, starting at `goal_onset_s`:

```
theta_goal(t) = A ∗ (1 − exp(−(t − t_onset)/tau_goal))    (t > t_onset)
F_goal = P(wrap(theta_goal − psi)) + (dtheta_goal − omega) ∗ V(...)
F_bar_dot = (−F_bar + F_feedback + ff_gain ∗ F_goal) / tau_bar
domega_ec/dt = (−beta ∗ omega_ec + F_bar + ff_gain ∗ F_goal) / I   (graded goal)
```

where `psi` and `omega` are the current proprioceptive values by default. For gate goal, `k_opto` is set to zero while `|F_bar + ff_gain ∗ F_goal| > gate_threshold`. Two ablations are available: `goal_delay_s` delays the goal trajectory itself, and `ff_self_state = delayed` replaces `psi` and `omega` in `F_goal` with the same delayed values used by the visual path. More generally, `ff_self_state_lag_s` ages the self-state sample by any amount, `ff_self_state = predicted` replaces the sample with a second-order Taylor reconstruction from the delayed observation alone (delayed states plus the delayed angular acceleration), and `ff_mode = damping` replaces the goal-tracking torque with the no-goal control `F_goal = −ff_damp_gain ∗ omega`, applied identically to the bar pathway, the efference copy, and the gate. The goal is instructed and shares its amplitude with the bar stimulus, so this branch is an upper bound on goal-based feedforward, not a biological claim; the onset is decoupled in one control.

For traceability, the manuscript names map to code and run-artifact names as follows. The code-level names are historical; the manuscript names are the ones to use in prose.

Manuscript name	Code model name	Run-summary label
addition	`addition`	`addition`
graded filtered	`graded_sensory`	`graded filtered`
graded command	`graded_command`	`graded command`
graded goal	`graded_goal`	`graded goal`
graded unfiltered oracle	`graded_motor`	`graded unfiltered oracle`
gate filtered	`allnone`	`gate filtered`
gate command	`gate_command`	`gate command`
gate goal	`gate_goal`	`gate goal`
gate unfiltered oracle	`allnone_motor`	`gate unfiltered oracle`

The unfiltered variants compute their bar torque from the current visual state; they are oracle controls, not motor signals.

The source's fixed graded implementation uses a gain of 0.05 on the bar torque in its Figure 5 code. We use 0.05 as the reference gain for the inert regime. The adaptive variant follows the source's `sim_effc` architecture with a five-unit sigmoid MLP. The weights are updated at each simulation step; a second implementation integrates them as ODE states with rate `alpha/dt`, and the two forms were compared pairwise over four delays, two update rules, and twelve parameter sets (96 pairs), agreeing on failure classification and failure time in every pair.

Delay

The delay is implemented by evaluating the visual state at `t - d` for the bar angle, grating angle, heading, and angular velocity. The body state and the torque filters are evaluated at the current time. The delayed values are stored in a ring buffer at the integration step; the delay is rounded to the nearest step. This is a transport delay, not a distributed latency.

Stimuli

The bar is a sigmoidal target turn of 60 degrees starting at 0.5 s with slope 30. The grating is a 30-degree sigmoidal background turn in the opposite direction starting at 0.75 s. The stimuli are deterministic. Unless a sweep is listed as noiseless, Gaussian torque noise with standard deviation `1e-13 N m` is added to the angular acceleration. The noiseless sweeps are the P1/P2/P3 delay sweeps, the main gain sweep, the adaptive check, the adaptive learning-rate sweep, and the ODE-equivalence check. Every later robustness sweep (gain robustness, window, V-mode, gate threshold, command copy, feedforward and its controls, adaptive signed-gradient, multi-seed, fine and long delays, noise, phase diagram, and trajectories) includes the noise term. The noise sweep itself varies the standard deviation from 0 to `1e-11 N m`.

Failure criteria and windows

The model is integrated with forward Euler at `dt = 1e-4 s` for the main sweeps, and `dt = 2e-4 s` for the gate-threshold sweep; a step-halving check on representative conditions changed the gating variants by less than 0.3% and the fast-failing variants by 2% to 16% near divergence. A run is a failure if the absolute heading exceeds `2*pi` radians, the absolute angular velocity exceeds 100 rad/s, or the state becomes non-finite. We report failure at 2.5, 6, and 10 s windows to test whether late failures are truncated. The failure criterion itself is also a choice: on three representative parameter sets, reclassifying the 1 ms fine-grid data with an angular-velocity threshold of 50 rad/s instead of 100 rad/s moves the gating critical delay from 200 ms to 56 ms for `gate filtered` and from 110 ms to 58 ms for `gate unfiltered`, while the continuous models stay between 10 and 18 ms under every criterion. The ordering (continuous variants fail first, gating later) is criterion-robust; the numerical gating boundary is not. A classifier-robustness check (720 runs: five representative conditions, eight parameter sets, three noise levels, three seeds, two integration steps, 6 s window) compared this peak criterion against a sustained criterion requiring at least 10 ms of continuous threshold exceedance. The two criteria never disagreed on a single trajectory: failures either stayed below threshold or diverged through it, and the peak absolute heading among failures had a median of `4e12` rad. Label flips across seeds, noise levels, and integration steps localized entirely to `gate filtered` at 100 to 140 ms with drag `1e-11 N m s` and `k_opto 5e-11 N m`, where the perturbation changes whether the trajectory diverges rather than how the trajectory is labeled. Failure rates in this boundary region are therefore

ensemble quantities over seed, noise, and integration-step perturbations; we report them as ensemble rates and do not attach meaning to single-run labels there.

Sweeps

The gain sweep crossed five models, six efference-copy gains (0, 0.05, 0.25, 0.5, 0.75, 1.0), four delays (0.02, 0.05, 0.10, 0.20 s), and 12 parameter sets (1,440 runs). The window check crossed five models, five delays, two seeds, three windows, and eight parameter sets (1,200 runs). The V-mode check crossed five models, five delays, two seeds, three V treatments, and eight parameter sets at a 6 s window (1,200 runs). The gate-threshold check crossed two gating models, four thresholds, four delays, two seeds, two V treatments, and eight parameter sets at a 6 s window (1,024 runs). The adaptive check crossed three update rules, four delays, and 12 parameter sets at a 6 s window (144 runs). The command-copy check crossed seven models (no efference copy, graded filtered, graded command, graded unfiltered, gate filtered, gate command, gate unfiltered), five delays, two efference-copy gains (0.05, 1.0), and eight parameter sets at a 6 s window (560 runs). The feedforward check crossed 15 model-gain combinations, five delays, one seed, and eight parameter sets at a 6 s window (600 runs), followed by a 10 s window check (400 runs), a goal-mismatch grid over three amplitudes and three time constants (648 runs), a goal-delay control over four goal delays and two visual delays (256 runs), and a self-state ablation (96 runs). The multi-seed check re-ran eight discriminating model-gain combinations at gain 1.0 with three seeds (960 runs). The ODE-internal adaptive check paired the step-wise and ODE-internal weight updates over four delays, two update rules, and twelve parameter sets (96 pairs). Five further checks extend the delay axis and the stimulus ensemble: a 10 ms fine grid from 0 to 250 ms for four model-gain combinations (832 runs, 6 s window), a long-delay check at 250 to 500 ms for six combinations (240 runs, 10 s window), a noise sweep over four standard deviations (800 runs), a body-parameter phase diagram over four inertias, three drags, two grating gains, and eight delays (384 runs), and an example-trajectory export at 200 ms (four runs, 10 s window). Three proposition tests complete the package: a goal-onset decoupling sweep over five onsets (360 runs), a no-goal damping control over three gains and both feedforward families (192 runs), and a self-state freshness sweep over eight sample lags at two visual delays (272 runs); an equal-information observer baseline comparing current, Taylor-predicted, and delayed self-state (320 runs). The feedforward window check re-ran both the 6 s and the 10 s windows (200 runs each); the goal-delay control crossed four model-gain combinations with four goal delays and two visual delays. A classifier-robustness check crossed five representative conditions, eight parameter sets, three noise levels, three seeds, and two integration steps at a 6 s window (720 runs).

Statistics

The primary quantity is the failure rate at a given delay, computed over parameter sets and seeds. We report counts and rates. No inferential test is applied because the parameter grid is a design, not a random sample. The failure criterion is a threshold crossing on a finite window, not an asymptotic stability boundary. Near the stability boundary, failure rates are ensemble quantities over seed, noise, and integration-step perturbations, as quantified by the classifier-robustness check.

Code and data availability

All code is in `M2A-BRIDGE/06_JOINT/`. Outputs are in `runs/`. A public archival deposit with a persistent identifier is planned at submission (DOI pending repository deposit; owner action). No new biological data were generated. The equations are ported from the published flight-simulator code accompanying Canelo et al. (2024), at https://github.com/nisl-hyu/flightsim (file `Figure S3/FIGURE_S3_delayed_vs_nondelayed.m`, inspected read-only). The source repository has no license; only equations were ported.

Figure legends

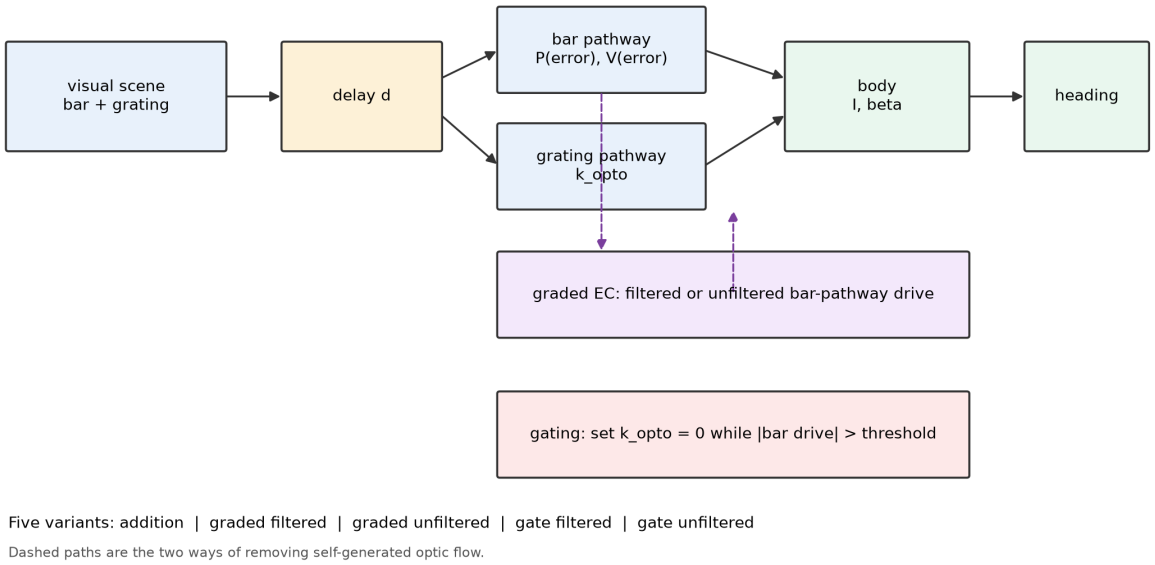


Fig. 1 | Model and delay manipulation. Schematic of the bar and grating pathways, the body, and the five variants. The delay is applied to the visually sensed state; the body and torque filters are current. Filtered and unfiltered refer to the bar-pathway signal used by the efference copy or gate, not to a sensory-versus-motor distinction.

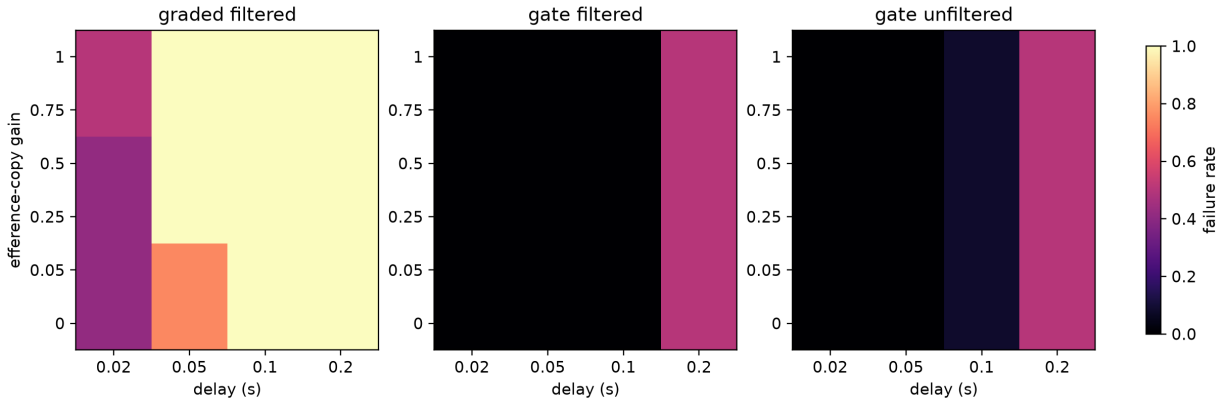


Fig. 2 | Efference-copy gain by delay. Failure rate for graded filtered, gate filtered, and gate unfiltered across efference-copy gains (0, 0.05, 0.25, 0.5, 0.75, 1.0) and delays (0.02, 0.05, 0.10, 0.20 s). At gain 0 or 0.05, graded filtered is indistinguishable from no efference copy. At gains of 0.25 or more, it is worse at 50 ms. Gating is gain-invariant by construction. Source: `fig2c_gain_delay_heatmap.png`.

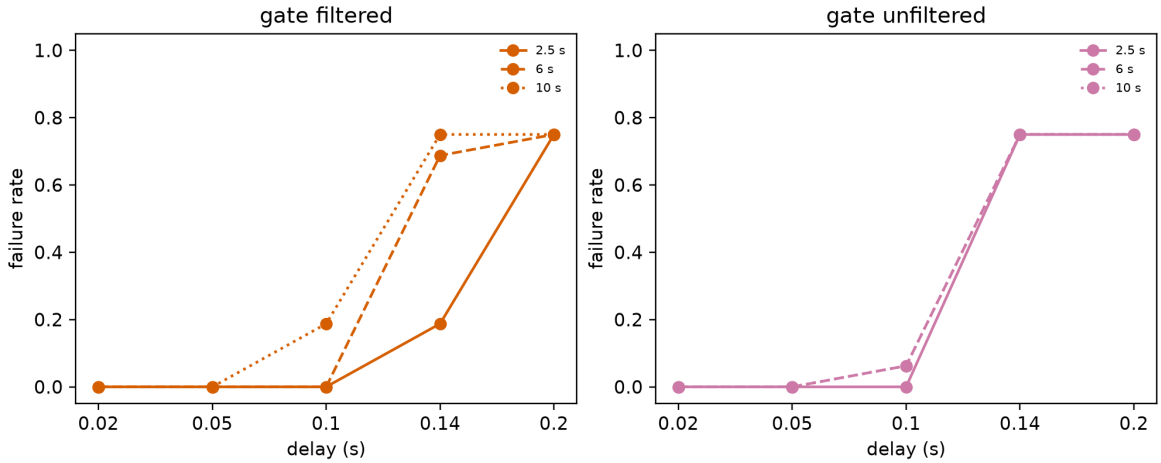


Fig. 3 | Window dependence of the gating comparison. Failure rate for gate filtered (left) and gate unfiltered (right) at 2.5, 6, and 10 s windows. At 140 ms, the 2.5 s window undercounts failures; at 6 and 10 s the two gating variants converge. Source: `fig_window_check.png`.

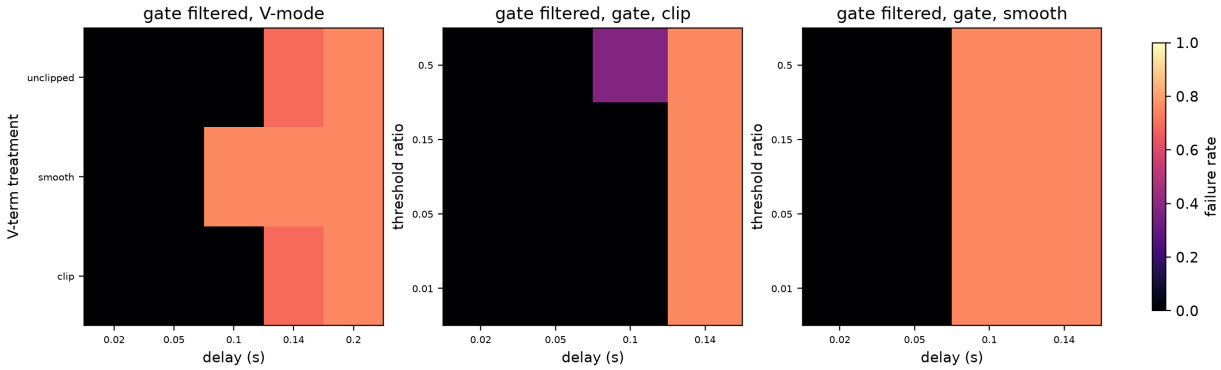


Fig. 4 | V-term and gate-threshold stress tests. Left, failure rate for gate filtered across clip, smooth, and unclipped V treatments. Middle and right, failure rate for gate filtered across gate thresholds from 0.01 to 0.5 of the bar-torque scale under clip and smooth V. The short-delay (20 to 50 ms) gating advantage for gate filtered survives all treatments; the 100 ms advantage does not. The second gating signal in the threshold sweep (gate unfiltered) matches these zeros except at threshold 0.5 under clip, where it fails in 25% of runs at 50 ms and 62.5% at 100 ms; that cell is not plotted here. Source: `fig_stress.png`.

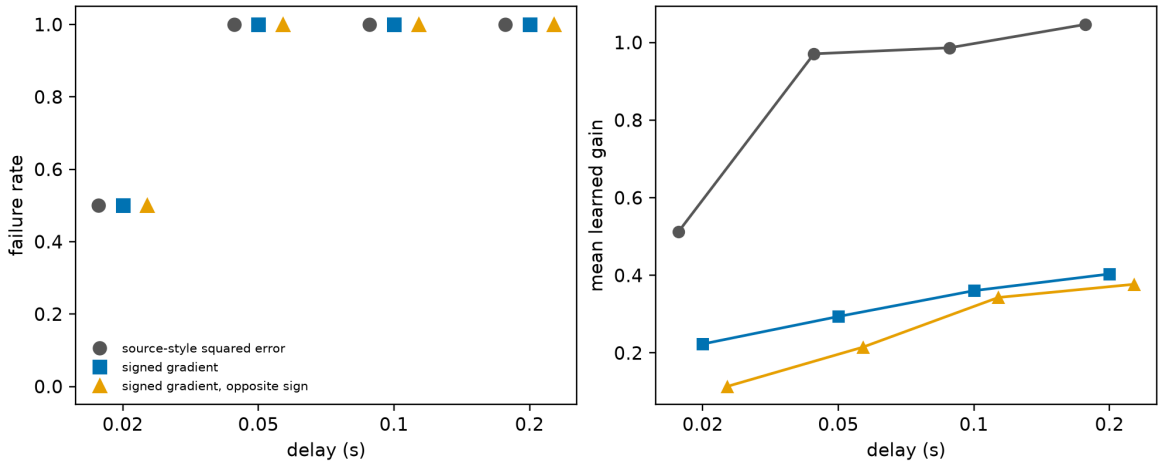


Fig. 5 | Adaptive updates. Failure rate (left) and mean learned gain (right) for the source-style squared-error update, a signed-error gradient, and the opposite-sign control. The signed update keeps the gain lower but does not change failure.

Source: `fig_adaptive_gradient.png`.

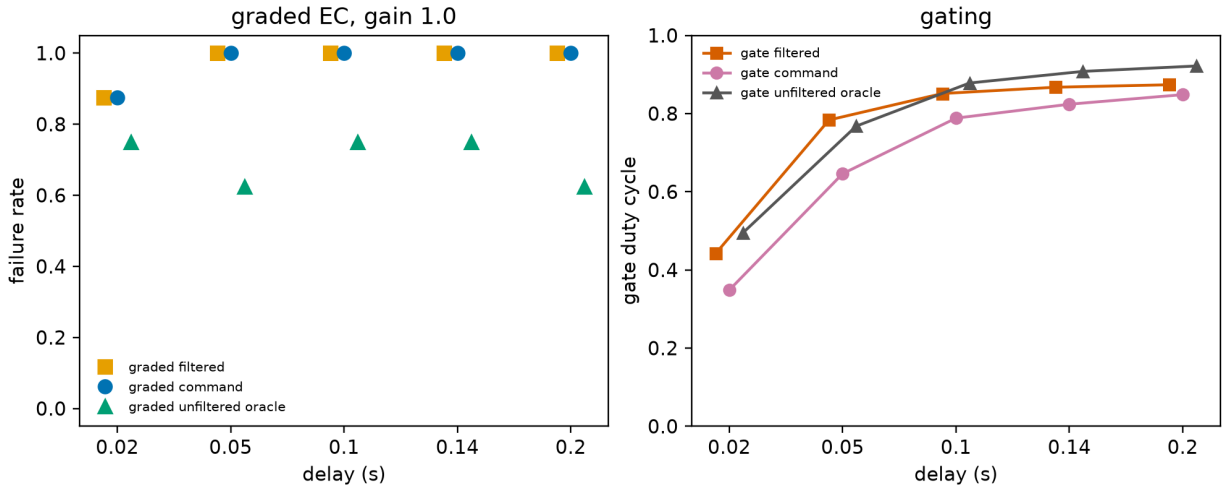


Fig. 6 | Command copy versus filtered plant state. Left, failure rate for graded efference copy at gain 1.0 for the filtered plant state, the pre-plant-filter command copy, and the current-visual oracle. Right, gate duty cycle for the same three signals; failure rates for the three gates were identical. The command copy overlaps the filtered plant state. Only the current-visual oracle differs. Source: `fig_command_copy.png`.

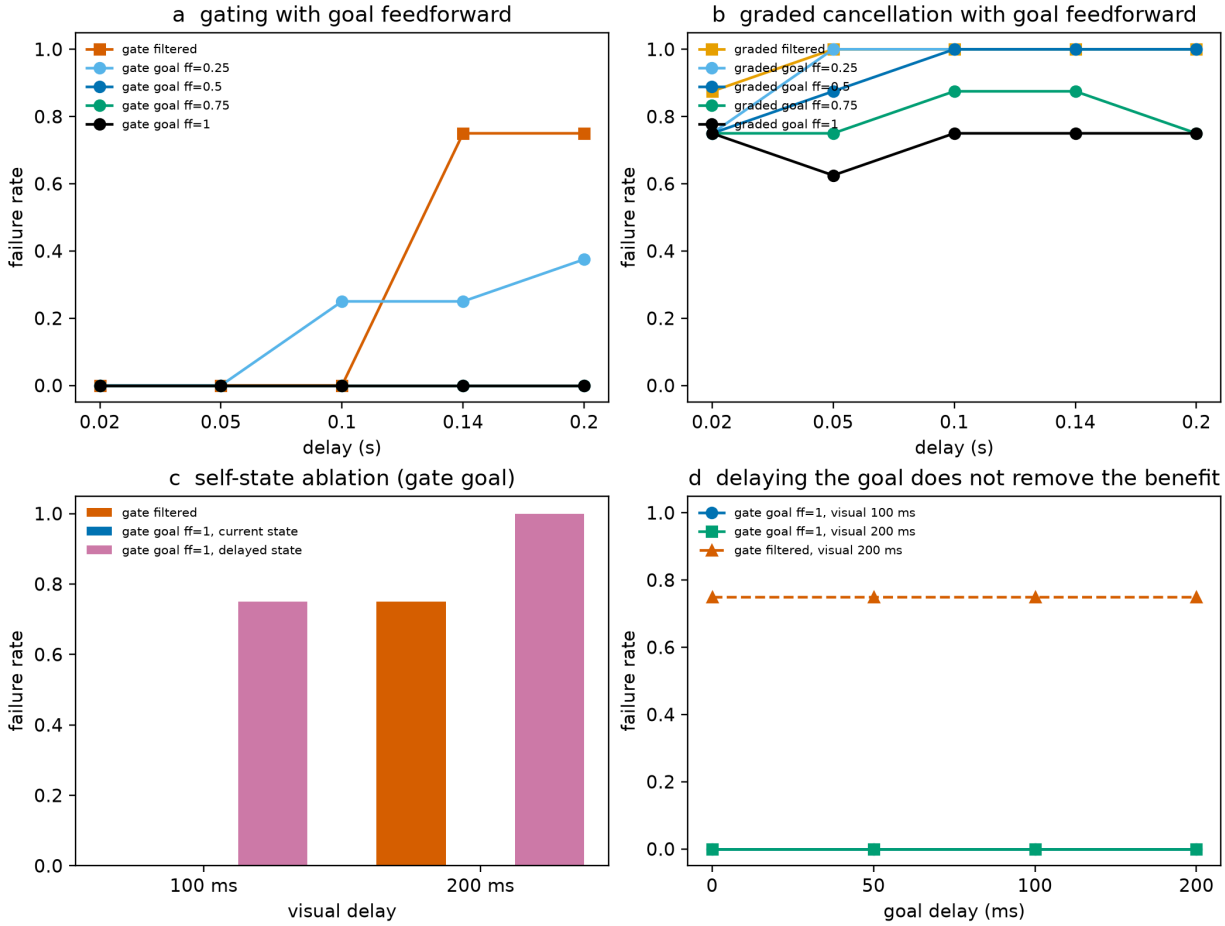


Fig. 7 | Goal feedforward and its mechanism controls. a, failure rate for gate filtered and gate goal across feedforward gains at a 6 s window. b, the same for graded cancellation. c, self-state ablation at 100 and 200 ms: gate goal at `ff_gain = 1.0` with current proprioception, with delayed self-state, and plain gate filtered. d, delaying the goal itself by 0 to 200

ms does not change failure; the visual delay does. Source: `fig_feedforward.png`.

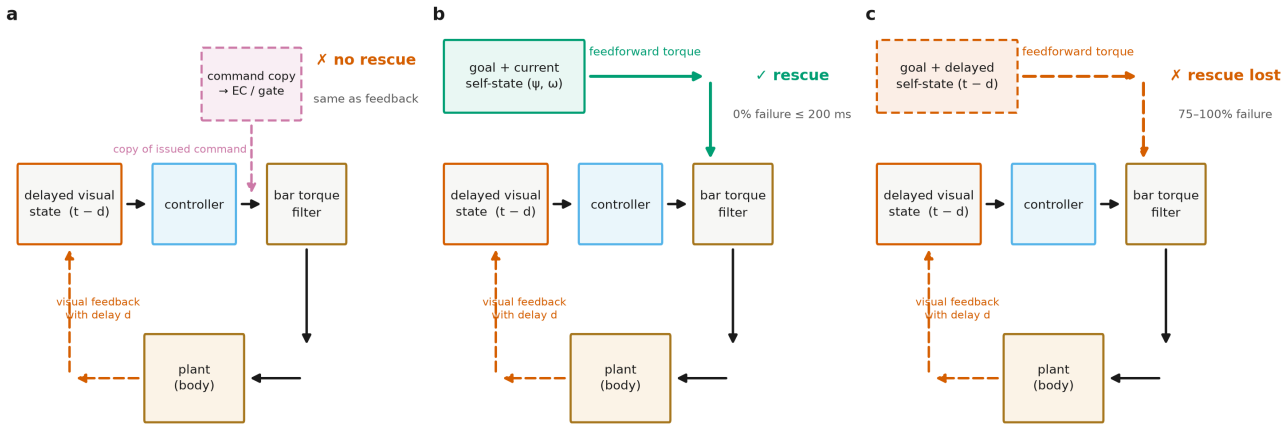


Fig. 8 | Mechanism summary. a, a command copy computed from the issued command adds no information beyond the delayed visual estimate and does not rescue. b, a goal feedforward torque computed from the instructed goal and current proprioception feeds the bar pathway and fully rescues through 200 ms and beyond. c, replacing current self-state with delayed values in that branch removes the rescue. Source: `fig_mechanism.png`.

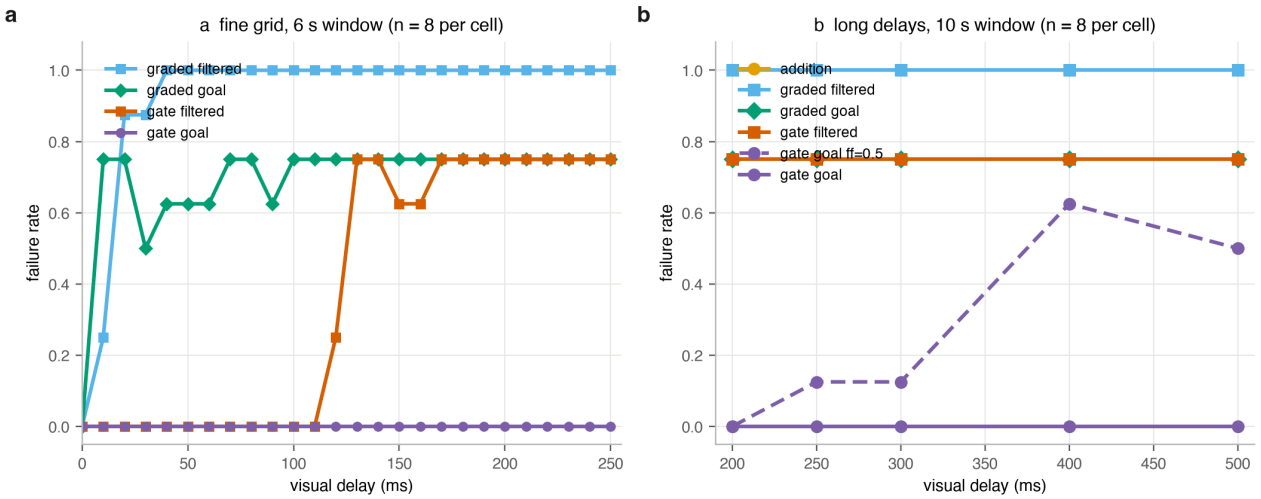


Fig. 9 | Delay limits. a, failure rate on a 10 ms grid from 0 to 250 ms at a 6 s window (n = 8 per cell). Gate filtered first crosses 50% failure at 130 ms; gate goal at `ff_gain = 1.0` never does. b, long delays at a 10 s window (n = 8 per cell). Gate goal at 1.0 stays at 0% through 500 ms; gate goal at 0.5 degrades beyond 200 ms. Source: `fig_delay_limits.png`.

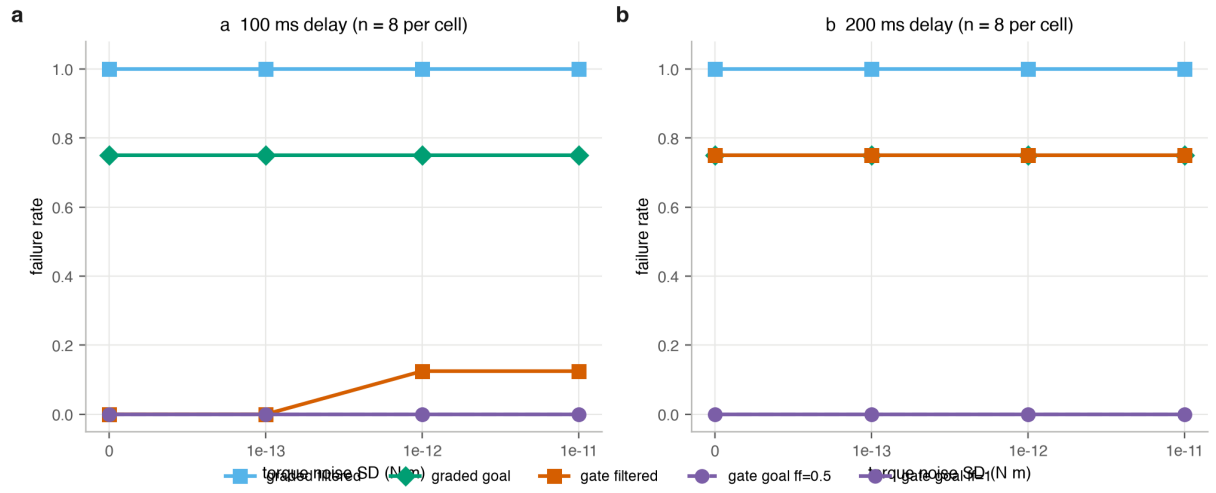


Fig. 10 | Noise robustness. Failure rate at 100 ms (a) and 200 ms (b) as the torque-noise standard deviation rises from 0 to `1e−11 N m` (n = 8 per cell). Gate goal at 0.5 and 1.0 stays at 0% at every noise level. Source: `fig_noise.png`.

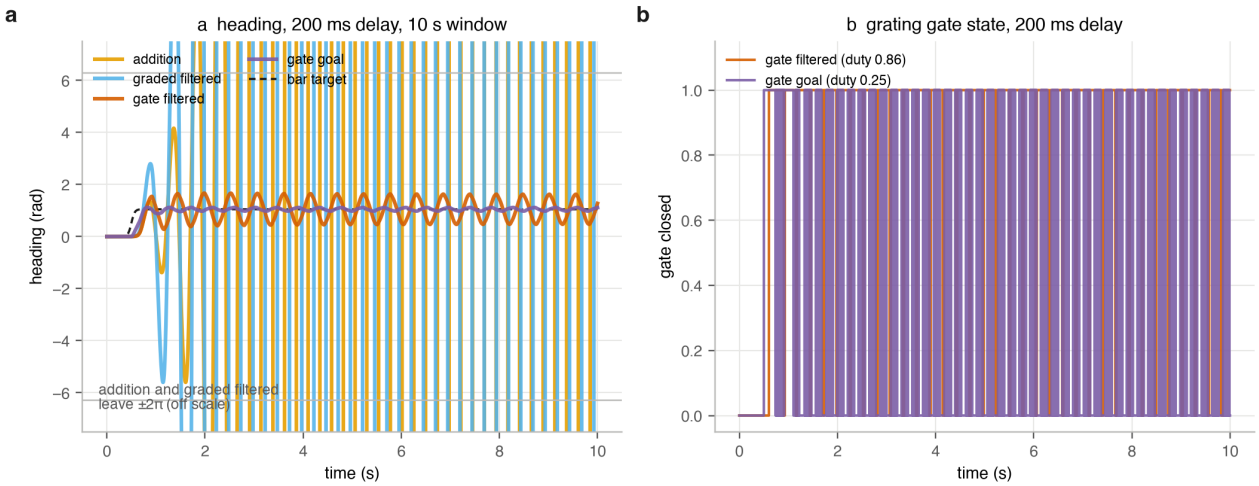


Fig. 11 | Example trajectories at 200 ms, 10 s window. a, heading for four variants with the bar target (dashed). Addition and graded filtered leave ±2π (off scale); both gates stay bounded. b, grating-gate state for the two gating variants; the filtered gate is closed in 86% of samples, the goal gate in 25%. Single parameter set and seed; illustrative, not statistical. Source: `fig_trajectories.png`.

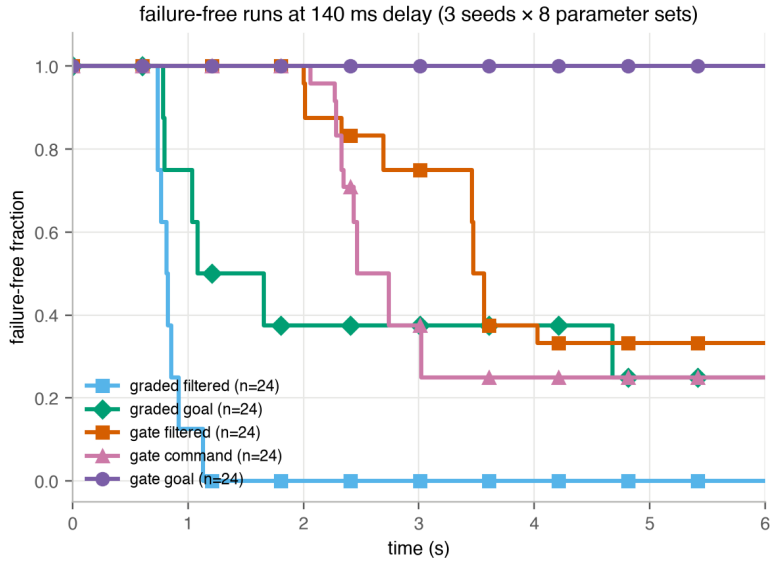


Fig. 12 | Failure-free curves at 140 ms. Fraction of runs without a failure over time, pooled over three seeds and eight parameter sets (n = 24 per curve) from the multi-seed check. Graded variants fail within the first second; gate goal never fails. Source: `fig_survival.png`.

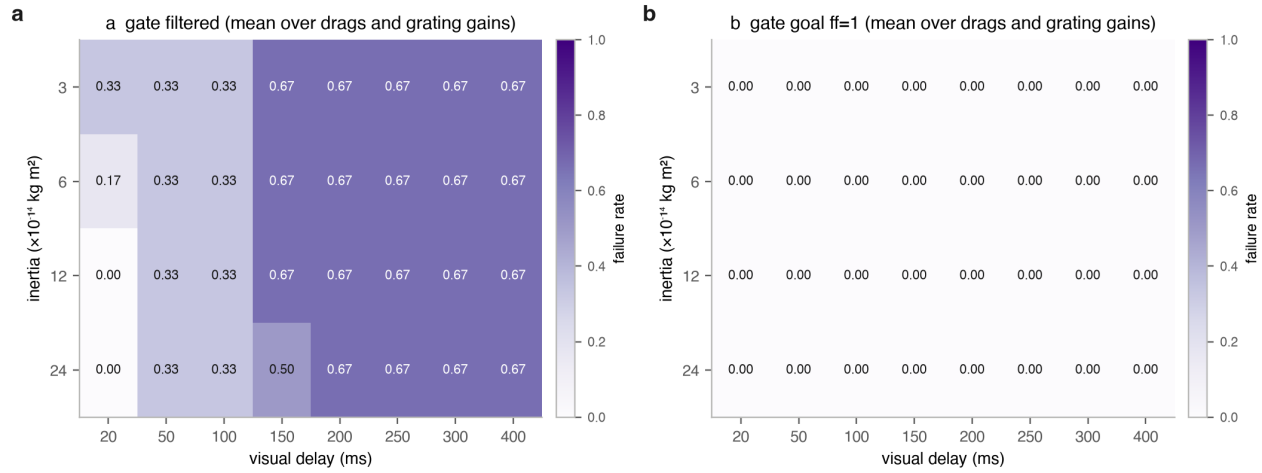


Fig. 13 | Body-parameter phase diagram. Failure rate averaged over drags and grating gains for gate filtered (a) and gate goal at `ff_gain = 1.0` (b), across four inertias and eight delays (12 body-gain combinations per cell). Gate goal fails in no combination at any delay; gate filtered degrades with body time constant and delay. Source: `fig_phase.png`.

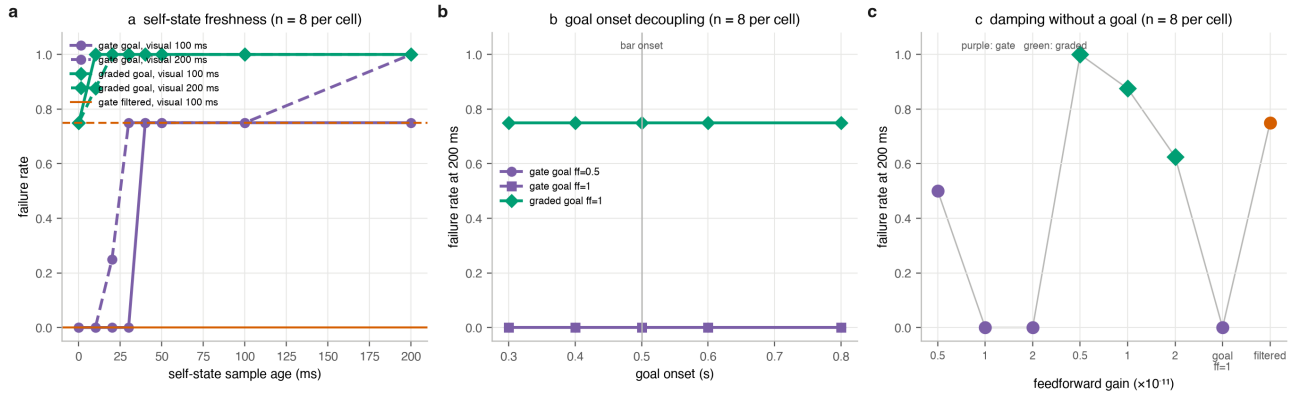


Fig. 14 | Freshness, timing, and content of the self-state signal. a, failure rate as the self-state sample ages, at visual

delays of 100 ms (solid) and 200 ms (dashed); gate goal degrades between 20 and 40 ms of sample age, graded goal by 10 to 20 ms; horizontal lines mark gate filtered. b, failure rate at 200 ms versus goal onset (bar onset fixed at 0.5 s, vertical line); no dependence for any variant. c, failure rate at 200 ms for the no-goal damping control across three gains (purple, gate; green, graded), with the goal-tracking gate and gate filtered as references; damping without any goal content rescues the gate. n = 8 per cell. Source: `fig_selfstate.png`.

Limitations

- The model is a toy model. It is not connectome-constrained and does not include the fly's actual visual circuit.
- Parameters are taken from Canelo et al. (2024) and not fitted to new data.
- The delay is a simple transport delay. Real sensorimotor delays are distributed and state-dependent.
- The failure criterion is a modeling choice.
- The filtered, unfiltered, and command variants differ in timing and filtering. The command copy is pre-plant-filter but still based on the delayed visual estimate.
- The observer baseline tests only a trivial Taylor estimator; a perfect internal-model observer is an untested upper bound, so "not a generic information-availability advantage" is established only relative to simple estimators.
- The goal feedforward is an upper bound, not a mechanism claim. The goal is instructed and shares its amplitude with the bar stimulus, and current proprioception is assumed available at a lag of at most a few tens of milliseconds (the freshness sweep puts the usable range at roughly 20 to 30 ms). The onset is decoupled in one control and does not matter. The feedforward branch changes the bar pathway itself, so it is a new pathway rather than a variant of the source model, and the same augmented signal drives both the corrective torque and the gate trigger. The damping control shows the rescue needs no goal content: a generic zero-delay self-state feedback signal suffices.
- The adaptive variant uses an oracle bar-only target; its step-wise weight update was verified to match an ODE-internal form (96/96 failure agreement), but the source's exact solver semantics are not reproduced.
- The model does not reproduce the source's zero-delay ranking. The source result is motivation, not validation.
- No new biological data. The predictions are untested.
- The model does not claim that the fly uses any of the model variants.
- The linear stability analysis was attempted but not used because the delay-eigenvalue tracking was unreliable.

References

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Scope note

This is a computational study. It does not report a new biological mechanism, a new animal experiment, or a new connectome. It does not claim that the fly uses any of the model variants. It does not claim that graded cancellation is biologically impossible. It is a model-based argument that efference-copy gain and short-delay gating are robust variables in this assay, that long-delay gating claims were artifacts of a finite window and a V-term clip, that a command copy alone does not rescue delay, and that a goal feedforward rescues only when it has access to current self-state.